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Insights into the physicochemical properties of newly synthesized benzyl triethylammonium chloride-based deep eutectic solvents

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ABSTRACT

In order to utilize deep eutectic solvents (DESs) as a replacement for conventional organic solvents for different applications, a detailed understanding of their physicochemical properties is extremely important. In this study, several novel DESs based on benzyl-triethylammonium chloride (BTEA) as a hydrogen bond acceptor (HBA) and acids/or alcohols as hydrogen bond donors (HBDs) have been synthesized. DESs have been thoroughly characterized and investigated for their physicochemical properties viz. thermal stability, glass transition temperature, viscosity (η), surface tension (γ), refractive index (n_D), density (ρ), speed of sound (u), and conductivity (κ) in a temperature range of 293.15–343.15 K. The effect of temperature, type of HBDs, and alkyl chain length of HBDs of the DESs on the physicochemical properties are critically examined and discussed. Viscosity data are fitted to Vogel-Fulcher-Tamman (VFT) and Arrhenius models. A linear model was used to fit surface tension (γ), density, refractive index, and speed of sound data whereas conductivity data were fitted to a nonlinear equation. Primary data on physicochemical properties have been used to estimate the viscous flow parameters (activation entropy, activation enthalpy, and free energy of activation) and other derived parameters such as lattice energy, standard entropy, molar volume, intermolecular free length, and isentropic compressibility. A relationship between molar conductivity and dynamic viscosity has been established to evaluate the ionicity of the synthesized DESs. Insights into the structure–property correlations are discussed in detail.

1. Introduction

Organic solvents, which are highly combustible, poisonous, and hazardous to the environment as well as to human health, are one of the main and most frequent sources of chemicals and industrial processes. To meet the requirements of green chemistry principles, new ecologically friendly and sustainable solvents must be developed [1]. Deep eutectic solvents (DESs) are well known for being the new adaptable, environmentally friendly, and designer solvents that can be easily created by merely combining the HBDs and HBAs in a specific molar ratio. The melting temperature of the generated liquid will be lesser than the melting points of their individual HBAs and HBDs. Abbott et al. published DESs for the very first time by mixing choline chloride (ChCl) as (HBA) and urea as (HBD) in 2003 [2]. The development of DESs depends on several interactions, including hydrogen bonding, van der Waals interaction, non-symmetricity, and mixing entropy. Even at high temperatures, several of the DESs combinations show chemical and thermal stability, with no chemical changes in their properties [3]. DESs can overcome the limitations associated with organic solvents by possessing good thermal stability, low vapor pressure, non-flammability, large liquidus range, and biodegradability [4]. Although DESs are sometimes referred to as a class of ionic liquids, they differ from ionic liquids in some ways, including being less hazardous, not requiring purification, and being simpler to synthesize, which makes them easier to scale up for large-scale manufacturing [5]. DESs can have either a hydrophilic or a hydrophobic character, depending on the makeup of the constituent parts. Long-chain alkyl groups typically cause an increase in hydrophobicity. For a long time, the potential of hydrophilic DESs has been investigated in a variety of disciplines, including biomass extraction, and processing, catalysis, enzyme activation, nanomaterials synthesis, gas separation, electrodeposition, battery technology, and pharmaceutical, etc [6–11]. However, the majority of hydrophobic DESs

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applications began in 2015, including the extraction of the malaria medication artemisinin from the leaves of *Artemisia Annua*, the first microextraction [12], extractions and separation of metals [13], photon up-conversion, and in membranes science are a few highly intriguing applications that have been studied [14].

When designing process equipment and scaling it up, an engineer must have a thorough understanding of thermophysical parameters connected to the solvent, such as density, viscosity, and refractive index. Pumping, mixing, and equipment size are all impacted by these characteristics, as well as mass and heat transmission. Data on thermophysical properties are also useful in choosing solvents that are appropriate for a certain procedure [15]. To better understand these characteristics of DESs, the scientific community is contributing its finest work. Accordingly, Soares et al. [16] investigated the density, viscosity, and Kamlet-Taft parameters of sodium hexanoate and decanoate-based eutectic solvents at a temperature range of 293.15–353.15 K. In the same temperature range of 293.15–343.15 K, Sánchez et al. [17] tested several additional physical characteristics, including refractive index, speed of sound, thermal expansion coefficient, isentropic-compressibility of proline and betaine-based DESs. To comprehend the fundamentals better, Achkar et al. [18] successfully wrapped up a discussion of several of these physical characteristics of DESs systems. Abbott et al. [2] measured the viscosity, refractive index, surface tension, and conductivity of [Ch]Cl: urea, and those of [Ch][Cl]: mono (phenylacetic, phenyl propionic), di (oxalic, malonic, adipic, and succinic), and tri (citric) acids [19], as well as the density, viscosity, conductivity, and surface tensions of [Ch]: ethylene glycol, glycerol, 1,4 butane diol [20]. The DESs [Ch]Cl: ethylene glycol and [Ch]Cl: glycerol, and their aqueous solutions, were measured between 298.15 and 333.15 K by Leron et al. [21] To evaluate the densities of nine distinct salts, Shahbaz et al. [22] utilized various HBAs such as methyl triphenylphosphonium, cholinium, and N-N, diethyl ethanolammonium, with numerous HBDs, including ethylene glycol, 2,2,2-trifluoroacetamide, and glycerol. It is also important to emphasize the work of Biernacki et al. [23]. They studied the type III DESs composed of xylitol, and sorbitol with choline chloride via experimental and computational approaches. In temperatures ranging from 293.15 to 353.15 K, researchers have studied the physicochemical characteristics of these DESs, including their density, viscosity, electrical conductivity, molar volume, refractive index, and molar refraction. Gardas and colleagues designed a new series of DESs by selecting benzyl tripropylammonium as HBA with lactic acid, phenol, glycerol, and ethylene glycol as HBDs. Several distinct thermodynamic parameters such as density, speed of sound, conductivity, and refractive index have been determined in the temperature range of 293.15 K–343.15 K [24]. In their expanded work reports, the same research group provide a new series of DES by replacing their previous HBA with current benzyl trimethylammonium chloride and benzyl tributylammonium chloride as HBAs with previously used HBDs. They carefully evaluated the characteristics of each DESs, such as density, sound speed, viscosity, refractive index, and electrical conductivity, and found that the molar ratio of HBD/HBA significantly affected the physicochemical characteristics of DESs in the temperature range of 283.15–343.15 K [25].

Herein, we have performed synthesis, characterization, and detailed physicochemical study of these newly synthesized DESs utilizing benzyl triethylammonium chloride (BTEA) as HBA with various organic acids (formic, acetic, butyric, valeric, pivalic, caprylic, capric, lauric, oleic, lactic, and levulinic acid) and alcohols (propane 1,2 diol, butane 1, 4 diol, and ethylene glycol) as HBDs. Investigations have been done into how several conditions, including temperature, alkyl chain length, and HBD type affect these physical properties. Physicochemical insights have been gained through precise measurements of primary physical properties such as viscosity, surface tension, refractive index, density, speed of sound, conductivity, and derived properties such as activation parameters of viscous flow (ΔH^*, ΔS^*, and ΔG^*), molar volume, lattice energy, standard entropy, intermolecular free length, and isentropic compressibility as a function of temperature (293.15–343.15) at atmospheric pressure. Correlations of viscosity and conductivity have also been evaluated and presented.

2. Experimental section

2.1. Materials

All the DESs were produced using benzyltriethyl ammonium chloride as the HBA in combination with a variety of HBDs, including propane 1,2, diol, butane 1,4 diol, ethylene glycol, lactic acid, levulinic acid, formic acid, acetic acid, butyric acid, valeric acid, pivalic acid, capric acid, caprylic acid, lauric acid, and oleic acid. Table 1 provides every detail pertaining to the substances employed in the current study. All chemical reagents were used as such without any additional sanitization. Fig. 1 displayed the chemical structure of the components of DESs.

2.2. Methods

2.2.1. Synthesis of deep eutectic solvents

The following HBDs such as oleic acid, lauric acid, caprylic acid, capric acid, pivalic acid, butyric acid, valeric acid, acetic acid, formic acid, lactic acid, levulinic acid, propane1,2-diol, butane1,4-diol, and ethylene glycol were mixed with the HBA like benzyl triethylammonium chloride (BTEA) in a particular molar ratio of 2:1 except pivalic for which ratio was (3:1) at 80.0 °C until they become a homogeneous low viscosity liquid, as shown in Fig. S1. Table 2 lists the synthesized DESs with the proper molar ratio used in this investigation. Synthesized DESs have been characterized using ^{1}H NMR spectra, Fig. S2 (a–n). The DESs designated as BTEA: OLE, BTEA: LAU, BTEA: CA, and BTEA: CAP are hydrophobic, whereas the remaining DESs are hydrophilic. The precaution was taken to prevent the DESs from absorbing any moisture in

Table 1
Details of chemical used in this study along with CAS number, supplier, and purity.

S. No.	Chemical name	CAS number	Supplier	Purity (as mass fraction)
1.	Benzyl triethylammonium chloride	56-37-1	TCI Chemical India, Pvt. Ltd.	≥98 %
2.	Lactic acid	50-21-5	TCI Chemical India, Pvt. Ltd.	≥85 %
3.	Levulinic acid	123-76-2	TCI Chemical India, Pvt. Ltd.	≥97 %
4.	Propane 1,2 diol	57-55-6	TCI Chemical India, Pvt. Ltd.	≥99 %
5.	Butane 1,4 diol	110-63-4	TCI Chemical India, Pvt. Ltd.	≥99 %
6.	Ethylene glycol	107-21-1	TCI Chemical India, Pvt. Ltd.	≥99.5 %
7.	Oleic acid	112-80-1	TCI Chemical India, Pvt. Ltd.	≥99 %
8.	Formic acid	64-18-6	Loba Chemie, India, Pvt. Ltd.	≥98 %
9.	Acetic acid	64-19-7	Loba Chemie, India, Pvt. Ltd.	≥99.8 %
10.	Capric acid	334-48-5	Loba Chemie, India, Pvt. Ltd.	≥98 %
11.	Caprylic acid	124-07-2	Loba Chemie, India, Pvt. Ltd.	≥99 %
12.	Lauric acid	143-07-7	Loba Chemie, India, Pvt. Ltd.	≥99 %
13.	Butyric acid	107-92-6	CDH finechemicals, India, Pvt. Ltd.	≥99.5 %
14.	Valeric acid	109-52-4	CDH finechemicals, India, Pvt. Ltd.	≥98 %
15.	Pivalic acid	75-98-9	CDH finechemicals, India, Pvt. Ltd.	≥99 %

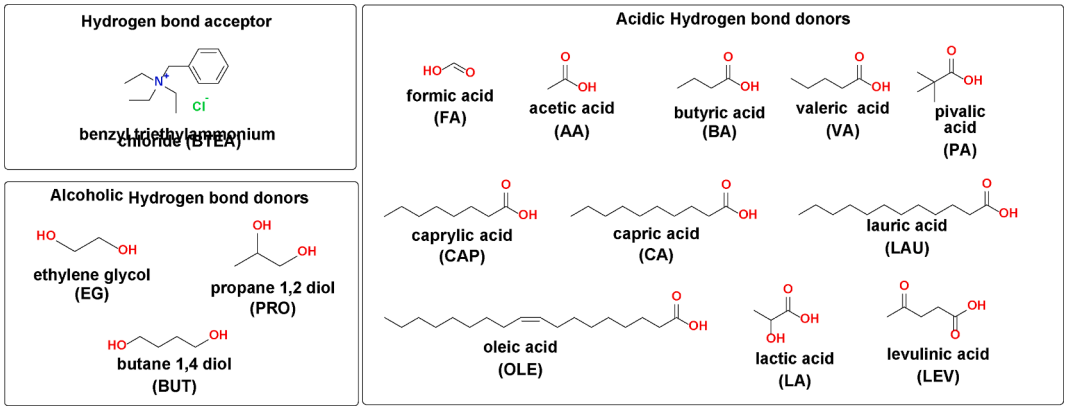


Fig. 1. The chemical structure of all DESs constituents.

Table 2
The abbreviations, components, molar ratio, mol wt, glass transition temperature (T_g), and decomposition temperature of DESs.[a]

Abbreviations	Component (HBA)	Component (HBD)	Molar ratio	Mol. Wt ($gmol^{-1}$)	T_g (K)	T_d (K)
BTEA: FA	Benzyl triethyl ammonium chloride	Formic acid	1:2	106.61	212.8	478.15
BTEA: AA	Benzyl triethyl ammonium chloride	Acetic acid	1:2	115.95	202.8	478.15
BTEA: BA	Benzyl triethyl ammonium chloride	Butyric acid	1:2	134.66	196.8	477.15
BTEA: VA	Benzyl triethyl ammonium chloride	Valeric acid	1:2	144.01	198.8	478.15
BTEA: PA	Benzyl triethyl ammonium chloride	Pivalic acid	1:3	133.54	211.8	473.15
BTEA: CAP	Benzyl triethyl ammonium chloride	Caprylic acid	1:2	172.06	196.8	453.15
BTEA: CA	Benzyl triethyl ammonium chloride	Capric acid	1:2	190.77	195.8	471.15
BTEA: LAU	Benzyl triethyl ammonium chloride	Lauric acid	1:2	209.47	192.8	478.15
BTEA: OLE	Benzyl triethyl ammonium chloride	Oleic acid	1:2	264.23	194.8	498.15
BTEA: LA	Benzyl triethyl ammonium chloride	Lactic acid	1:2	135.97	194.8	473.15
BTEA: LEV	Benzyl triethyl ammonium chloride	Levulinic acid	1:2	153.33	192.8	478.15
BTEA: PRO	Benzyl triethyl ammonium chloride	1,2 Propane diol	1:2	126.65	199.8	468.15
BTEA: BUT	Benzyl triethyl ammonium chloride	1,4 Butane diol	1:2	136.00	194.8	498.15
BTEA: EG	Benzyl triethyl ammonium chloride	Ethylene glycol	1:2	117.30	193.8	488.15

[a] Standard uncertainties in temperature, mole fraction, and pressure are $u(T) = 0.1$ K, $u(x) = 0.005$, and $u(P) = 1$ kPa respectively.

the air, and the produced DESs were subsequently stored in desiccators. The physical properties of the DESs were examined at various temperatures using advanced equipment and an average of triplicate measurements.

2.2.2. Physicochemical characterization

All the synthesized DESs underwent thermogravimetric analysis using the Mettler Toledo Star SW 8.10 system in a nitrogen environment and at a rate of 5 °C/min, and DSC analysis by Phoenix DSC 204 F1 instrument with 10 °C rise per minute. Viscosity measurements for all the synthesized DESs were carried out using an Anton Paar Dynamic Viscosity: AMVn Automated Micro Viscometer (relative standard uncertainty 0.01). The Attention Sigma 700 (relative standard uncertainty 0.1) was used to measure the surface tension. Anton Paar (Model DSA 5000) vibrating tube density meter was used to measure the density (relative standard uncertainty of 0.005), and speed of sound (relative standard uncertainty of 0.001). Refractive indices have been recorded using a Mettler-Toledo (Model RE-40 D) refractometer (relative standard uncertainty 0.0001) with a high-resolution optical sensor. Conductivity (κ) measurements were performed using a Thermo Scientific™ Eutech™ PC 2700 Multi-parameter Meter with pH Electrode (ECFG73711B), Conductivity Electrode (CONSEN921D), and ATC Probe (PH5TEMB1P) (relative standard uncertainty 0.02).

3. Result and discussion

3.1. Thermal Stability, and glass transition temperature determination

To ascertain the thermal stability of the synthesized DESs in terms of decomposition temperature, T_d, a thermogravimetric study was carried out. Fig. 2 shows the thermogram of newly synthesized DESs whereas the thermal stability of remaining earlier synthesized DESs is reported in our previous article [5]. Table 2 includes the associated degradation temperature for each DESs. It can be clearly observed that all the synthesized DESs show a good thermal stability range from 453.15 to 498.15 K. The observed thermal stability of DESs is showing comparable results with the previous findings [26]. Increasing chain length has been found to increase the viscosity, and DESs with higher viscosities have greater thermal stability [25]. Here, DESs with acids having comparatively smaller molecules such as BTEA: FA, BTEA: AA, BTEA: BA, BTEA: VA, and BTEA: PA has little effect on the thermal stability whereas, a significant increase in thermal stability has been observed for acids of significant chain length with increase in chain length such as BTEA: CAP (453.15 K), BTEA: CA (471.15 K), BTEA: LAU (478.15 K), and BTEA: OLE (498.15 K). This pattern is also seen in other DESs, such as BTEA: LA (473.15 K), BTEA: PRO (468.15 K), which exhibit less thermal stability than BTEA: LEV (478.15 K), and BTEA: BUT (498.15 K).

One of the most noteworthy qualities in DESs, in addition to the decomposition temperature, is the melting temperature, particularly when used to replace standard solvents. The range of temperatures where DESs can maintain their liquid state depends on these features. The melting point of the mixture has been consistently demonstrated to be lower than that of its constituent parts, making it impossible for DSC to detect most DESs. Instead, each DES displayed a glass transition temperature, proving that these DESs are supramolecular complexes capable of existing in a liquid state over a wide temperature range [45]. Additionally, the hydrogen bonds formed by the HBA and HBDs also serve to shield anion from cation, reducing electrostatic interaction and, ultimately, melting point [46]. The melting temperatures of various DESs, such as BTEA: PA, BTEA: CAP, BTEA: CA, BTEA: LAU, and BTEA:

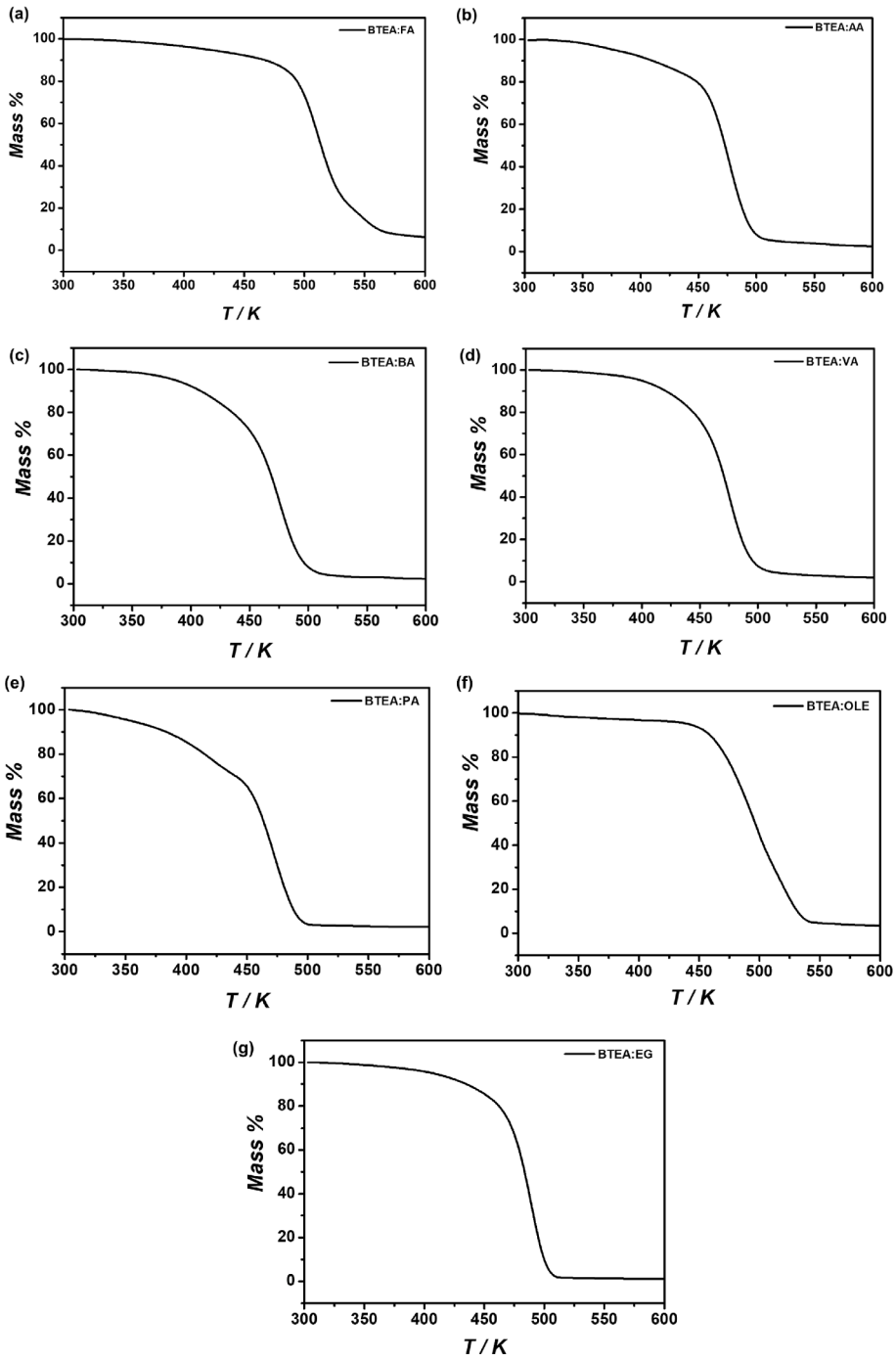


Fig. 2. The thermogravimetric analysis curve of synthesized DESs.

OLE, were found to be 275.79, 266.80, 288.79, 304.8, and 274.79 K by DSC. These obtained melting temperatures of DESs are also in good agreement with earlier reported DES made up of these components [12]. Importantly, these DESs also have a lower melting point than their initial constituent. The glass transition temperatures (T_g) discovered through this investigation are added to Table 2 and their corresponding plots are displayed in Fig. S3.

3.2. Viscosity

To describe the solvent's mobility and flow resistance, viscosity is an important physical property. It is also critical to research how temperature impacts DESs viscosity to judge their utility and calculate the energy required to process them [27]. Fig. 3a illustrates the experimental fluctuations in viscosities of all the synthesized DESs with temperatures ranging from 293.15 K to 343.15 K at 5 K intervals. The plot clearly shows that, as would be predicted, viscosities appear to be dropping exponentially as the temperature rises. The Hole Theory, which governs the mobility of species into mixtures, can be used to explain temperature effects. When the DESs mixture is heated, additional energy is added, which speeds up and spreads out molecules, creating large holes that allow molecules to pass each other with ease and subsequently reduce viscosity [28]. The experimental viscosity data of the DESs are summarized in Table S1. The Vogel-Fulcher-Tamman (VFT) model and Arrhenius model are the two models that are most frequently used to analyze how temperature affects viscosity in DESs.

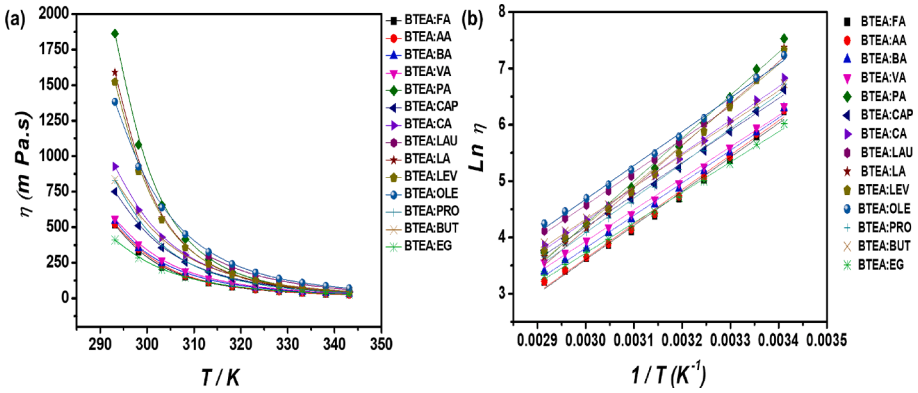


Fig. 3. (a) The plot of variation in viscosity (as per VFT equation (1) with rising temperature, (b) Arrhenius plot (equation (2) representing linear relation between ln n vs 1/T.

$$\eta = \eta^{o} exp\left(\frac{B}{T - T_O}\right) \tag{1}$$

Equation (1) represents the VFT equation, where the limiting value of viscosity is denoted by η^o at high temperature, B is the fitting parameter, and T_o is the Vogel temperature. After fitting the experimental viscosity data into the VFT equation, the value of the adjustable parameters and the regression coefficient are presented in Table S2. Additionally, the experimental data were also fitted in the logarithmic form of Arrhenius equation (2).

$$\ln\eta = \ln\eta^{O} + \frac{E_n}{RT} \tag{2}$$

Where ln η is the logarithm of viscosity, η^o is the pre-exponential factor and E_n is the activation energy for momentum transfer (kJ mol^{-1}). The obtained linear relation between viscosity and temperature has been displayed in Fig. 3b and the corresponding values of E_n, η^o, and R^2 for each prepared DES are listed in Table 3. With an R^2 value of 0.99, it was demonstrated that there was a substantial correlation between the temperature and viscosity of the investigated DESs. According to reports, the E_n values for ChCl: Urea (1:2) [29], ChCl: glycerol (1:2) [30], ChCl: malonic acid (1:1) [19], and BT: EG (1:3) [24] are 51.27 kJ mol^{-1}, 42.37 kJ mol^{-1}, 44.68 kJ mol^{-1}, and 44.90 kJ mol^{-1} respectively. The obtained E_n value listed in Table 3 has a good correlation with the body of available literature. Comparing these two models reveals that the VFT model offers a better fit ($R^2 > 0.999$) than the Arrhenius model ($R^2 >$ 0.994–0.997), demonstrating the viscoelastic nature of DESs. Many DESs have demonstrated this type of behavior in the past. For example, Cui et al. [31] found that the viscosity of choline chloride with *para*-toluene sulfonic acid, propaionic, mono, and trichloroacetic acid-based DESs fit the VFT more closely than Arrhenius. Like this, the VFT model is preferred to Arrhenius in another investigation of the viscosity of DESs based on benzyltrialkyl ammonium chloride [25].

Table 3
Value of adjustable parameters of viscosity with regression coefficient as per equation (2).

DESs	E_n/kJ/mol	n^0/mPa s	R^2
BTEA: FA	50.8401	4.71942E-07	0.992
BTEA: AA	50.3582	4.05247E-07	0.994
BTEA: BA	48.0172	1.35139E-06	0.996
BTEA: VA	46.4494	2.68098E-06	0.995
BTEA: PA	63.3235	8.26373E-09	0.993
BTEA: CAP	49.5723	2.2935E-06	0.996
BTEA: CA	48.5614	1.25216E-06	0.996
BTEA: LAU	47.6285	2.31903E-06	0.997
BTEA: LA	61.7892	1.31035E-08	0.992
BTEA: LEV	59.3057	3.44626E-08	0.991
BTEA: OLE	49.9148	1.61921E-06	0.997
BTEA: PRO	51.1929	5.62581E-07	0.995
BTEA: BUT	47.3998	2.8128E-06	0.997
BTEA: EG	44.7975	3.94485E-06	0.997

In addition to the impact of temperature, chain length has an important function. It was found that when the molecular weight of the various DESs increased, so did their viscosities. This pattern is consistent with the Mark-Houwink equation, which is frequently used to characterize the viscosity behavior of polymers [32]. Thus, as chain length rises, the molecular weight tends to rise as well, increasing viscosity. As a result, The BTEA: FA DES corresponds to lower viscosity among all the studied acidic DESs at lower (293.15 K, 514.469 mPa.s) as well as at higher temperatures (353.15 K, 24.625 mPa.s) due to its shorter chain length. Sarmad et al. [33] evaluated the viscosity of various DESs for screening towards green CO_2 sorbent. From there, our measured results are in good agreement with previously reported values for the dynamic viscosity of one of the DESs (BTEA: AA) that was the subject of this investigation. In addition to temperature and chain length, the viscosity is also dependent upon the nature of HBDs [34]. A similar tendency of increased viscosity has also been noted when acids are replaced by alcohol. Because of this BTEA: EG (411.478 mPa.s) is having a lower viscosity than BTEA: PRO (828.389 mPa.s), and BTEA: BUT (834.398 mPa.s) at 293.15 K.

3.2.1. *The association activation parameters studied for all DESs*

It is also important to define the operational settings for the process after acquiring experimental viscosity data. Then, because viscosity is a direct result of intermolecular interactions, it can offer information on a molecular basis. Determining thermodynamic activation parameters of viscous flow using the Eyring equation is one of the methods for understanding intermolecular interactions from viscosity [35]. The associated activation parameters for their viscous flow, namely ΔH^* (enthalpy of activation), ΔG^* (free energy of activation), and ΔS^* (entropy of activation), have been earlier analyzed for various ionic liquids [36,37]. Considering the energy of activation (ΔG^*), which is identical to the enthalpy of activation, (ΔH^*). The Arrhenius version of viscosity relations can be expressed as,

$$\eta = \left(\frac{hN_A}{V_m}\right) e^{\frac{\Delta H^*}{RT}} e^{\frac{-\Delta S^*}{R}} \tag{3}$$

Where h is Planck's constant in J.s, N_A is Avogadro's number (mol^{-1}), R is the gas constant (8.314 $JK^{-1}mol^{-1}$) and V_m is the molar volume in m^3 mol^{-1}. Taking the logarithm of the above equation, we get

$$\ln\eta = \left\{\ln\left(\frac{hN_A}{V_m}\right) - \frac{\Delta S^*}{R}\right\} + \frac{\Delta H^*}{RT} \tag{4}$$

To calculate the ΔG^* (kJ mol^{-1}) the above equation (4) can be written as

$$\Delta G^{*} = RT\left(\ln\frac{\eta V_m}{hN_A}\right) \quad (5)$$

The Gibbs-Helmholtz equation (6) is used to determine ΔH^* (kJ mol^{-1}).

$$\Delta H^{*} = \left(\frac{\partial(\Delta G^{*}/T)}{\partial(1/T)}\right)_P \quad (6)$$

The activation entropy (kJ K^{-1} mol^{-1}) is obtained by using equation (7).

$$\Delta S^{*} = \frac{\Delta H^{*} - \Delta G^{*}}{T} \quad (7)$$

The calculated values of activation parameters (ΔG^*, ΔH^*, and ΔS^*) is represented in Tables S3–S5. The behavior of these activation parameters concerning temperature is shown in Fig. 4. The process is assumed to be non-spontaneous since the viscous flow is confronted by the activation barrier, as shown by the positive values of ΔG^*. The Gibbs free energy values are falling with increasing temperature, indicating that less energy is required to initiate viscous flows as shown in Fig. 4a. It is interesting to note that as the chain length grows, more energy is needed to start the viscous flow. Because of this, BTEA: OLE (33.567 kJ mol^{-1}) has the highest ΔG^* value among acid-based DESs, whereas BTEA: FA (28.591 kJ mol^{-1}) has the lowest ΔG^* value at 293.15 K. Similarly, in DESs, based on alcohol, this rising trend of ΔG^* is also evident as BTEA: EG (28.303 kJ mol^{-1}) < BTEA: PRO (30.254 kJ mol^{-1}) < BTEA: BUT (30.460 kJ mol^{-1}). The process involved heat absorption, producing positive ΔH^*, which naturally decreased with rising temperature, confirming a decrease in the compactness of the DESs structure and a rise in supporting DESs mobility (Fig. 4b). For instance, BTEA: FA DESs has the highest activation enthalpy (65.353 kJ mol^{-1}) at 293.15 K and the lowest activation enthalpy (37.796 kJ mol^{-1}) at 343.15 K. Since the activation entropy values (ΔS^*) are positive, $\Delta G^* < \Delta H^*$ follows. Positive values imply that entropy rises after reaching the transition state, which frequently denotes a dissociative process in which the activated complex is loosely bound and on the verge of dissociating. Consequently, as the temperature increases from 293.15 K to 343.15 K, the activation entropy also decreases, as seen in Fig. 4c. Interestingly, the DESs investigated in this study and ILs investigated previously have positive but too much smaller activation entropy (<0.18 kJ K^{-1} mol^{-1}) values [37].

3.3. Surface tension

Surface tension is a practical physical characteristic that sheds light on the molecular ensembles and free ions in the liquid in terms of both short- and long-range forces. Free ion mobility and these forces can coexist in harmony due to the temperature, chain length, and nature of HBDs [38]. The development of novel separation technologies and the resolution of numerous industrial-related issues must be evaluated. Surface tension data for the DESs measured using the Du Noüy ring

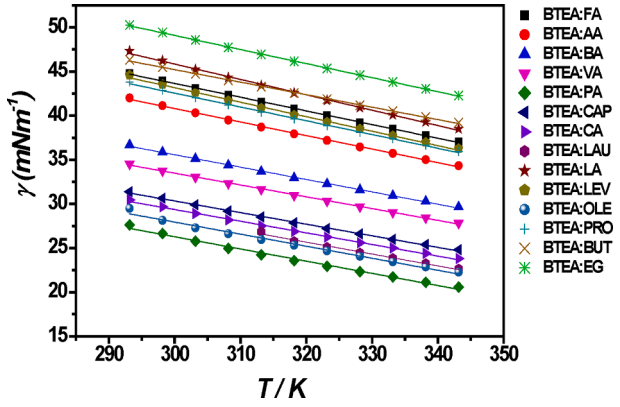


Fig. 5. The plot of surface tension of all synthesized DESs with temperature variation.

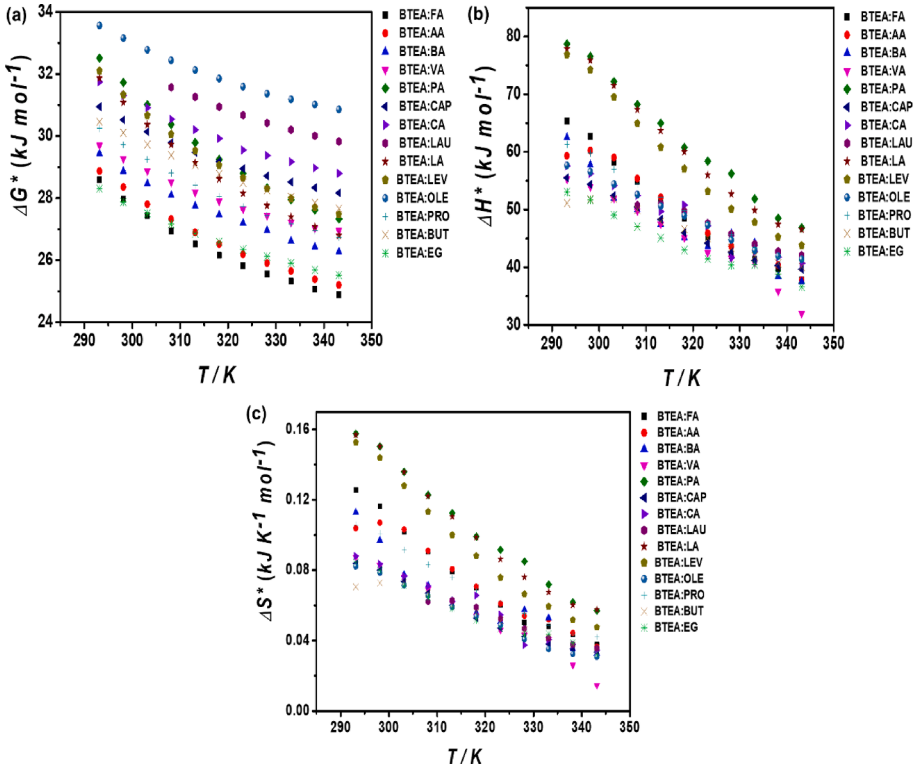


Fig. 4. The plot of derived viscous flow activation parameters concerning temperature, (a) ΔG^* (free energy of activation), (b) ΔH^* (enthalpy of activation), and (c) ΔS^* (entropy of activation).

method is provided in Table S6. Fig. 5 shows how the surface tension varies when the temperature rises from 293.15 K to 343.15 K.

The plot illustrates that the surface tension of DESs appears to decrease linearly with increasing temperature. The decrease in surface tension of all the synthesized DESs may be attributed to the reduction in the binding force between the DES elements by the increased thermal energy of the molecule with temperature. As a result, surface tension loses some of its cohesive power. The nature and alkyl chain length of the HBDs also maintains the balance between such forces and free ion movement. Surface tension increases with decreasing alkyl chain length [39,40]. Due to this, BTEA: FA exhibits the highest surface tension of all acid-based DESs, except for BTEA: LA and BTEA: LEV, which comes from different classes of acids. The order for surface tension is followed as BTEA: FA > BTEA: AA > BTEA: BA > BTEA: VA > BTEA: CAP > BTEA: CA > BTEA: LAU, and BTEA: OLE. By the substitution of alcohols for acids as the HBDs. It was interesting to see that the surface tension of DESs containing alcohol was higher than that of DESs based on acids, which may be related to the strong hydrogen-bonding interaction between HBA and alcohols. The surface tension is often reduced by the weaker hydrogen-bonding interaction, whereas the surface tension is increased by the hydrogen-bonding network [41]. Hence, BTEA: EG exhibits the highest surface tension of all the DESs under study. The variation in surface tension is influenced by component type, mole ratio, electrostatic, van der Waals, and intermolecular hydrogen bonding interactions [39]. Previously reported surface tension data of similar kinds of DESs (having the same HBDs) such as ChCl: LA, 1:2 (47.4 mNm^{-1}), ChCl: EG, 1:2 (49.4 mNm^{-1}), TEAC: AA, 1:2 (40.9 mNm^{-1}) [41], closely matches with investigated DESs; BTEA: LA, 1:2 (47.3 mNm^{-1}), BTEA: EG, 1:2 (50.2 mNm^{-1}), and BTEA: AA, 1:2 (42.0 mNm^{-1}). A linear model equation (8) has been used to fit the obtained surface tension (γ) data, and the resulting parameters, a and b with R^2 are displayed in Table 4 which demonstrated a linear relationship between the temperature and DESs surface tension. Such a linear correlation between surface tension and temperature of ammonium/phosphonium-based DESs has also been observed by Shahbaz et al. [42] In the fitted equation (8) slope b might be used to describe the rate at which surface tension decreases as a function of temperature. Since lowering surface tension as the temperature rises results in a constant negative slope value, it is implied by a larger negative value of the slope that the surface tension is more susceptible to changes in temperature affecting the rate at which its surface tension decreases. The slope and intercept value of all DESs also decreases with increasing chain length as depicted in Table 4.

$$\gamma = a + bT \tag{8}$$

3.4. Refractive index

Following Snell's law, the refractive index of a substance gauges how much the speed of light alters as it passes through a medium in comparison to the speed of light in a vacuum as per equation (9).

$$n_D = c/\nu \tag{9}$$

The nature of DESs, the type of HBD, temperature, and molecular weight of DES are some of the variables that might affect the refractive index of DESs. We have measured the refractive indices of all the synthesized DESs under the temperature range 293.15 K–343.15 K at 5 K intervals. Fig. 6 shows the variation of the refractive index with temperature. The experimental refractive index data are shown in Table S7. The obtained results reveal that for all the synthesized DESs there is a linear decrease with the temperature rise. The strength of the interactions between molecules affects the refractive index. When the temperature rises, molecules move more quickly, which reduces the strength of their interactions. In other words, when the temperature rises, the liquid loses density and viscosity, allowing light to move through the medium more quickly. Light moves more slowly through a medium at a lower temperature because the liquid gets thicker and more viscous [17]. The effect of the alkyl chain length of the HBDs is such that, the longer the alkyl chain the smaller the n_D values. As a result, at 293.15 K BTEA: FA DES is showing a larger refractive index value among all acid-based DESs. When it comes to alcohol-based DESs, the pattern is still the same, with BTEA: EG (1.5169) having the highest value, followed by BTEA: PRO (1.5114), and BTEA: BUT (1.5087). Like how BTEA: LA (1.5074) has a higher value than BTEA: LEV (1.5017), it dominates. The experimentally determined refractive index value agrees with the findings of Basaiahgari et al. [24].

The experimental refractive index data were well fitted to equation (10), and the adjustable parameters (a and b), combined with the R^2 value listed in Table 5, indicated that the linear equation provided the best match.

$$n_D = a + bT \tag{10}$$

Analyzing the nature of the system is aided by knowing how n_D and other thermophysical properties relate to one another. The refractive index gauges a substance's dielectric reaction to an applied external electromagnetic field (light). Lorenz's first-order approximation adequately captures the relationship between density and refractive index. Inferring from the Lorentz relation, free molar volume (F_m) in liquids can be deduced by using the following equations (equations (11), 12) [42].

$$R_m = \frac{M(n_D^2 - 1)}{\rho(n_D^2 + 1)} \tag{11}$$

$$F_m = V_m - R_m \tag{12}$$

where n_D is the refractive index, R_m is the molar refractivity, ρ is the density, and M is the molar mass. In the Lorenz-Lorentz approximation,

Table 4
Surface Tension fitting parameters for all synthesized DESs.

DESs	a	b	R^2
BTEA: FA	90.1807	−0.1551	0.999
BTEA: AA	86.7462	−0.1531	0.998
BTEA: BA	77.6476	−0.1402	0.998
BTEA: VA	73.5774	−0.1336	0.998
BTEA: PA	67.7618	−0.1382	0.991
BTEA: CAP	69.7622	−0.1314	0.998
BTEA: CA	68.9360	−0.1319	0.997
BTEA: LAU	68.8037	−0.1347	0.987
BTEA: LA	98.1234	−0.1742	0.997
BTEA: LEV	92.4282	−0.1641	0.996
BTEA: OLE	68.9079	−0.1365	0.986
BTEA: PRO	89.4097	−0.1562	0.998
BTEA: BUT	87.7109	−0.1416	0.998
BTEA: EG	96.9220	−0.1594	0.999

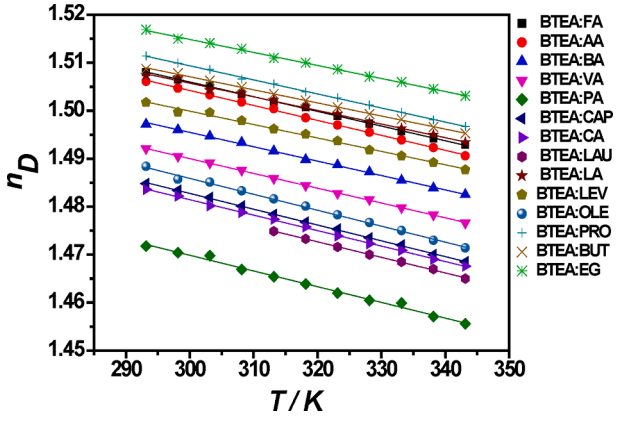


Fig. 6. The plot of variation in the refractive index of all DESs with increasing temperature.

Table 5
Value of adjustable parameters of refractive index with regression coefficient.

DESs	a	b × 10^4	R^2
BTEA: FA	1.5977	−3.06	0.999
BTEA: AA	1.5976	−3.11	0.999
BTEA: BA	1.5853	−3.00	0.998
BTEA: VA	1.5824	−3.08	0.999
BTEA: PA	1.5681	−3.27	0.992
BTEA: CAP	1.5815	−3.29	0.999
BTEA: CA	1.5781	−3.22	0.999
BTEA: LAU	1.5764	−3.24	0.998
BTEA: LA	1.5909	−2.84	0.994
BTEA: LEV	1.5837	−2.79	0.994
BTEA: OLE	1.5855	−3.32	0.996
BTEA: PRO	1.5972	−2.93	0.998
BTEA: BUT	1.5889	−2.73	0.999
BTEA: EG	1.5962	−2.71	0.998

R_m values from the equation above are considered as the hard-core molecular volume. As a result, it is assumed in many liquid-state models that the molecular structure contains free volumes, also known as free molar volume (F_m), and is not filled by any molecules. Simple subtraction of the R_m from the molar volume (equation (12) yields this result. The experimental data and graphs related to R_m and F_m are represented in Fig. S4, Table S8, and Fig. S5, Table S9. Data shows that there is no appreciable variation in R_m values with temperature change. Despite the fact that a rising temperature was associated with an increasing trend in both R_m and F_m. It is significant to note that the values of R_m and F_m are directly correlated with chain length, which accounts for why BTEA: FA has lower R_m (37.537 $cm^3.mol^{-1}$) and F_m (58.926 $cm^3.mol^{-1}$) values than BTEA: OLE (104.532, 172.021 $cm^3.mol^{-1}$). The value of R_m is presumably dependent on the same parameter as V_m, where a decrease in density causes an increase in both R_m and V_m. However, the refractive index, an additional factor in R_m, attempts to lower R_m as the temperature rises. Because of this, the increase in R_m value was not as great as the increase in V_m. This means that the primary cause of the rise in F_m is the increase in V_m. These volumes have important uses for screening a solvent for gas solubility, according to Hu et al [43].

3.5. Density

How temperature can influence DESs density is often required in establishing their industrial applications [44]. Therefore, the densities of all the synthesized DESs were measured between the temperature range of 293.15 K–343.15 K at 5 K intervals. The experimental data have been listed in Table S10. Fig. 7 displays the variations in DESs densities as a function of temperature for all DESs, which declined linearly as temperature increased. This occurs because when a liquid or gas is heated, the molecules move more quickly, collide with one another, and disperse apart. The molecules require a larger space since they are distributed widely. Thus, an increase in temperature provides thermal energy to the DESs constituents which in turn results in the increase of free volume of the systems and thereby decrease in density. The physical characteristics of DESs can be significantly influenced by the length of the alkyl chain. The free volume rises and the density falls as the hydrocarbon chain length increases [45]. Based on the Hole theory, local DESs density variations caused by thermal energy might result in rising space within the DESs. Due to the expansion of the hydrocarbon chains, which results in an increase in free volume and a fall in DESs densities [46]. BTEA: FA (1.1052 g cm^{-3}) DES has the highest density when compared to other DESs made from comparable non-substituted organic acids due to its least carbon content. The density order follows in this manner BTEA: FA > BTEA: AA > BTEA: BA > BTEA: VA > BTEA: PA > BTEA: CAP > BTEA: CA > BTEA: LAU > BTEA: OLE. In comparison to carbonyl-containing acids like BTEA: LA (1.1360 g cm^{-3}) has a higher density than BTEA: LEV (1.1148 g cm^{-3}). Betaine with lactic acid (BLA, 1:2) and levulinic acid (1:2) have similar density values of 1.2014 and 1.1583 g cm^{-3}, respectively at 298.15 K according to Sánchez et al [17]. The density result was obtained by Sethi et al [47]. on choline chloride and acetic acid (1.1106 $g.cm^{-3}$) based DESs likewise best match our result as our BTEA: AA shown 1.0824 $g.cm^{-3}$ at 298.15 K. Greater chain length in our DES system helps to explain this little discrepancy. The same phenomenon was also observed in alcohol-based DESs such as (BTEA: PRO (1.0685 g cm^{-3}), BTEA: BUT (1.0621 g cm^{-3}), and BTEA: EG (1.0945 g cm^{-3}). Another interesting point to be noted here is that hydrophobic DESs (BTEA: CAP 0.9585, BTEA: CA 0.9426, BTEA: LAU 0.9271, and BTEA: OLE 0.9234 g cm^{-3}) exhibited lower density value than water (0.9772 g cm^{-3}), whereas hydrophilic DESs had a higher density than water at 343.15 K. The densities of benzyl trimethyl and benzyl tributyl ammonium chloride using ethylene glycol-based DESs were 1.1039 g cm^{-3} and 1.0390 g cm^{-3}, respectively, according to Gardas and their coworkers [25]. It is interesting to observe that the density of benzyl triethyl ammonium chloride in ethylene glycol is reported here as BTEA: EG (1.0945 g cm^{-3}), which substantially follows the diminishing trend with increasing length from methyl to ethyl to butyl. Similar observations can also be seen in the work of Canales and their coworkers [48].

The DESs densities were fitted using a linear equation (13).

$$\rho = a + bT \tag{13}$$

Where ρ corresponds to density, T is for temperature, and a, and b are the DESs dependent constants. The resulting fitting parameters have been shown in Table 6 which suggested a good correlation between temperature with density. As can be observed, the value of both parameters (a, b) reduced as chain length increased (without considering the negative sign), which might be related to a decline in hydrogen bond interaction [49]. The obtained slope (b) value of all synthesized DESs in

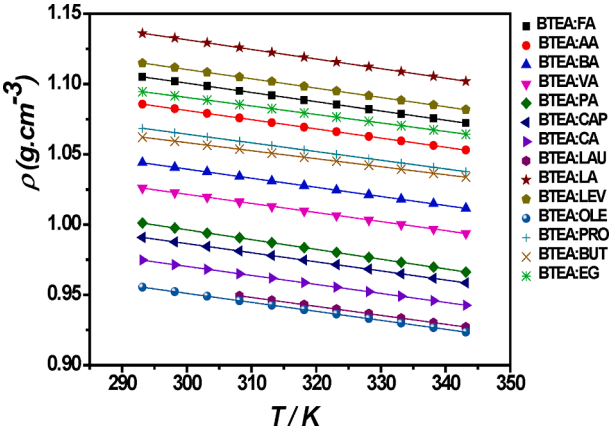


Fig. 7. The plot of variation in density of all DESs with increasing temperature.

Table 6
Value of adjustable parameters of the density of synthesized DESs.

DESs	a	b × 10^4	R^2
BTEA: FA	1.29822	−6.59	0.999
BTEA: AA	1.27749	−6.54	0.999
BTEA: BA	1.23420	−6.49	0.999
BTEA: VA	1.21534	−6.48	0.999
BTEA: PA	1.20514	−6.96	0.999
BTEA: CAP	1.18079	−6.44	1
BTEA: CA	1.16243	−6.40	1
BTEA: LAU	1.14560	−6.38	1
BTEA: LA	1.33641	−6.83	0.999
BTEA: LEV	1.30950	−6.64	0.999
BTEA: OLE	1.14280	−6.36	1
BTEA: PRO	1.24961	−6.08	1
BTEA: BUT	1.22919	−5.70	1
BTEA: EG	1.27179	−6.14	1

this work is showing a good correlation to previously reported DESs systems [50]. It was intriguing to observe that the single acid-based DESs (described here) displayed a lower density value than the double acid-based DESs published by Florindo et al. [45] This phenomenon might occur because double acids interact with hydrogen bonds more frequently.

3.6. Speed of sound

To determine the essential thermodynamic characteristics of DESs for use in chemical and industrial processes, measurements of the speed of sound (u) are an effective parameter. Table S11 and Fig. 8 contain the experimental data, and variation of the speed of sound for all the synthesized DESs from 293.15 K to 343.15 K at 5 K intervals. From Fig. 8 we can see that the speed of sound of all the DESs is linearly decreasing with the increase in temperature. Speed of sound is regarded as a thermodynamic property that is directly related to density such that, as there is a decrease in the density of the medium concerning the rise of temperature, the speed of sound of all the DESs tends to get decreased. The medium used to carry sound waves becomes less dense as a result of the absorption of greater thermal energy at higher temperatures, which in turn slows down the speed of sound [50]. The effect of alkyl chain length also follows the same order as density.

The empirical relation developed by Jacobson [51] for determining intermolecular free length (L_f) as follows (Equation (14) can be used to further validate this quality.

$$L_f = K\sqrt{\frac{1}{\rho u^2}} \tag{14}$$

$$\beta_s = \frac{1}{\rho u^2} \tag{15}$$

where K, also known as Jacobson's constant, is a temperature-dependent variable, ρ (g cm^{-3}) is the density, and u (ms^{-1}) is the speed of sound, according to the computed values of L_f at atmospheric pressure in Table S12, it rises as the temperature rises. The slower sound speed at higher temperatures is a result of greater space. An essential thermodynamic parameter, the isentropic compressibility (βs), can be derived using the well-known Newton-Laplace equation [52] as follows Equation (15). The amount of free space that is present in the liquid can also be determined by looking at the isentropic compressibility. The estimated values shown in Table S13 show that the βs rise as the temperature rises. The obtained βs value of benzyl triethylammonium chloride-based DESs is showing a good correlation with previously observed benzyl tripropylammonium chloride-based DESs [24]. Fig. 9a, b represents the variation in intermolecular free length and isentropic compressibility concerning temperature.

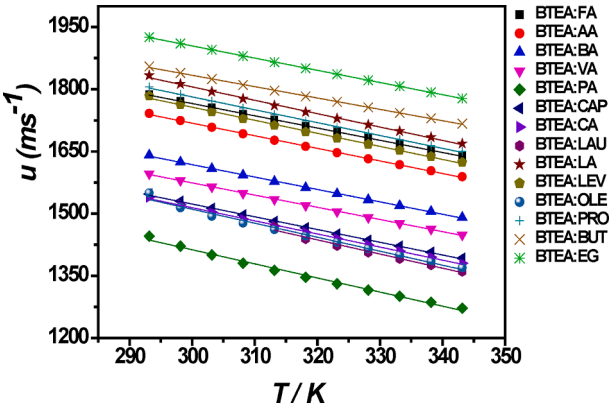


Fig. 8. The plot of variation in the speed of sound of DESs with increasing temperature.

3.7. Molar volume

The molar volume (V_m) for all the synthesized DESs has been derived from the density values. At a specific temperature and pressure, it is the volume that one mole of a substance (DESs component) occupies. As stated by Abbott et al. [53] Equation (16) was used to determine the molar volume (V_m) under typical pressure and temperature conditions.

$$V_m = \frac{x_{HBA}M_{HBA} + x_{HBD}M_{HBD}}{\rho} \tag{16}$$

where M represents the molar mass ($g.mol^{-1}$), ρ is the density (g. cm^{-3}), x is the mole fraction, and V_m is the molar volume (in $cm^3.mol^{-1}$) respectively. The calculated molar volume data of all the synthesized DESs is displayed in Table S14 and Fig. 10a. The values of the derived molar volumes appear to rise as the temperature rises. As can be seen in Fig. 10a the inverse relationship between density and molar volume exists. So as the temperature increases the free volume of the system increases leading to an increase in the molar volume and a decrease in density. At 293.15 K, increasing chain length also results in a rise in molar volume that ranges from 96.46 $cm^3.mol^{-1}$ (lowest for BTEA: FA) to 276.55 $cm^3.mol^{-1}$ (highest for BTEA: OLE). Similarly, from BTEA: LA (119.69 $cm^3.mol^{-1}$) to BTEA: LEV (137.54 $cm^3.mol^{-1}$), the rise in molar volume has also been noted. As with alcohols based DESs, the molar volume rises from 107.16 to 118.53 to 128.04 $cm^3.mol^{-1}$ as the chain lengthens from BTEA: EG to BTEA: PRO to BTEA: BUT. Again, at 293.15 K the molar volume of our BTEA: EG (107.16 $cm^3.mol^{-1}$) DES lie between that of benzyl trimethyl ammonium chloride and benzyl tributyl ammonium chloride with ethylene glycol-based DESs as of 84.45 $cm^3.mol^{-1}$ and 120.22 $cm^3.mol^{-1}$ supports the fact that lengthening the chain length increases the molar volume. According to equation (16), growing the chain length increases the molecular weight of the DESs, which raises the molar volume.

3.8. Standard entropy

The amount of entropy in one mole of a pure substance at any desired temperature and a standard condition of pressure is known as standard entropy. The sign so denotes a substance's standard entropy, indicating that it was calculated under standard conditions. At zero degrees, all molecular motion stops (0 K). As a result, zero is the definition of the entropy of a pure crystalline solid at absolute zero. Using the equation, the standard entropies of all the synthesized DESs were determined (equation (17).

$$S^0 = 1246.5(V) + 29.5 \tag{17}$$

where S^0 is the standard entropy in $JK^{-1}mol^{-1}$ and V is the molecular volume in nm^3 ($V = 10^{21} * V_m/N_A$) [54]. As predicted, it was discovered that the estimated values increased with the increase in temperature, much like molar volume as can be observed in Fig. 10b and Table S15. It has to do with the fact that as the temperature of substances rises, their entropy does also due to an increase in molecular mobility. Additionally, we can conclude that standard entropy has a direct relationship with molar volume and an inverse relationship with density. Since standard entropy and molar volume have a direct relationship, the molecular weight of DESs affects its value in a similar way to how molar volume does. The standard entropy, therefore, grows as chain length does. At 293.15 K, The BTEA: FA (229.1 $JK^{-1}mol^{-1}$) has lower standard entropy than all other DESs based on comparable acids. In comparison to the other two alcohol-based DESs, BTEA: EG (251.3 $JK^{-1}mol^{-1}$) is less entropy-intensive than BTEA: PRO (274.8 $JK^{-1}mol^{-1}$), and BTEA: BUT (294.5 $JK^{-1}mol^{-1}$), and likewise BTEA: LA (277.2 $JK^{-1}mol^{-1}$) is having less standard entropy than BTEA: LEV (314.2 $JK^{-1}mol^{-1}$). It has been found that regardless of the nature or kind of DES, any examined DESs with the same molar volume will have the same standard entropy. As an illustration, Nowosielski et al. [55] published the standard entropy

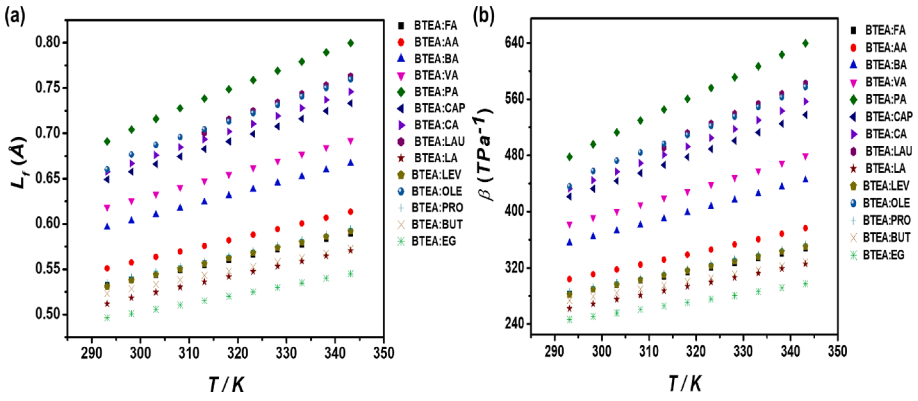


Fig. 9. The plot of intermolecular free length (a) and isentropic compressibility (b) of all DESs with temperature variation.

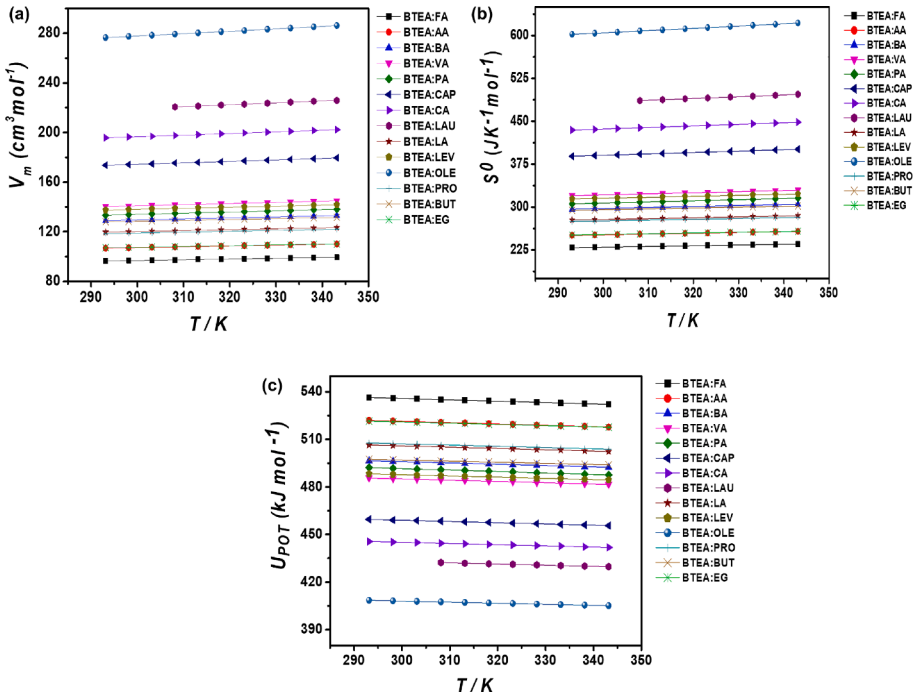


Fig. 10. The variation in molar volume (a), standard entropy (b), and lattice energy (c) of all DESs concerning temperature variation.

value of DESs made of 3-amino-1-propanol (AP) with three different ammonium salts, including tetrabutylammonium bromide, chloride, and tetraethylammonium chloride. According to their report, it can be noted that tetrabutylammonium bromide and chloride with 3-amino-1-propanol (AP) has a standard entropy of 281.8 and 279.6 $JK^{-1}mol^{-1}$ with a molar volume of around 121.64 $cm^3.mol^{-1}$. Similarly, in our example BTEA: LA has a standard entropy (277.9 $JK^{-1}mol^{-1}$) that is like the report mentioned above since it has the same molar volume (120.04 nm^3) at 298.15 K, and the subsequent DESs likewise follow this pattern.

3.9. Lattice energy

The lattice energy is the quantity of energy necessary to change one mole of an ionic solid into its gaseous ionic components. It depends on how strongly the ions in the system interact with one another. Using equation (18), which is based on the Glasser theory [56], one might compute the lattice energies of all the synthesized DESs.

$$U_{POT} = 1981.2(\rho/M)^{1/3} + 103.8 \quad (18)$$

where U_{POT} is the lattice energy in kJ mol^{-1}. The calculated lattice energy values for all the DESs were decreased with respect to the increase in temperature as shown in Fig. 10c and Table S16. The amount of charge carried by the DESs constituents and the space between them both have a significant impact on the lattice energy of a DESs combination. The distance between molecules in a lattice increases with temperature and decreases with the strength of the forces binding them together and hence, lattice energy [56]. The lattice energy is impacted by both the density and molecular weight of DESs components where density is directly proportional and molecular weight is inversely proportional to the lattice energy. However, the lattice energy is strongly influenced by molecular weight than density among all studied DESs as per equation (18). Additionally, comparing the lattice energy to the molar volume (according to equations (16), and 18) shows that it tends to be the reverse of the latter, with increasing molecular weight tending to increase molar volume in the former instance. As a result of molecular weight, the BTEA: FA DES has the highest (536.4 kJ mol^{-1}), and the BTEA: OLE has the lowest lattice energy (408.4 kJ mol^{-1}), at a temperature of 293.15 K. The calculated lattice energy of all DESs is having good similarity with the previous literature [25].

3.10. Conductivity

The interest of the research community in power system applications, including their possible usage in new electrolytes for redox-flow batteries, has been sparked by DESs conductivity [57]. Therefore, for every application, it is crucial to understand the conductivity of DESs. Accordingly, we have measured the variation of conductivity with temperature from the range 293.15 K to 343.15 K at 5 K intervals. Fig. 11a shows the variation of specific conductivities of all the synthesized DESs with increasing temperature. The experimental-specific conductivity data are listed in Table S17. The calculated molar conductivity (by equation (20) data and graph with temperature variation are listed in Table S18 and Fig. S6. As expected, we have observed from the data that the conductivity is showing an exponential increase with the increase in temperature. This is mainly due to the increase in the mobility of the molecules by the absorption of thermal energy with the temperature rise. The nature of DESs and molecular weight are the variables that affect their thermal conductivity. Since DESs and classical ILs share a similar chemical structure, it is hypothesized that these characteristics will likewise affect the conductivities of DESs [23]. The difference in the conductivities of the synthesized DESs is attributed to the chemical structure of the HBDs used, as all the DESs contain benzyl triethylammonium chloride as the HBA. As a result, it has been observed that the thermal conductivity of the corresponding DES falls as the molecular weight of the HBD grows. The larger alkyl chain length reduces the ability of molecules to migrate inside the components of DES, which lowers their conductivity. Because of this BTEA: OLE (23.27 μs cm^{-1}) DESs exhibited the lowest conductivity among all studied DESs at 293.15 K. Additionally, hydrophilic DESs showed greater conductivity than hydrophobic DESs (BTEA: CAP, BTEA: CA, BTEA: LAU, and BTEA: OLE). In alcohol-based DESs, BTEA: EG predominates with a higher conductivity than the other two, although BTEA: PRO and BTEA: BUT have a similar amount of conductivity at 293.15 K. The opposing tendency between conductivity and viscosity, which results in lower conductivity for greater viscosity DESs, is another crucial point to be made. This can also be understood in terms of magnitude and sign of the activation energy for conductivity and viscosity which are opposed to one another (see Tables 3 and 7). Since mass transport must overcome a large energy barrier, DESs with higher viscosity have lower conductivity, which has a lower equivalent activation energy [25].

The DESs conductivities at different temperatures were modeled to the Arrhenius equation (19).

$$\ln k = \ln k^{o} + \frac{E_a}{RT} \tag{19}$$

Table 7
Value of adjustable parameters of conductivity with regression coefficient.

DESs	(-) E_a (kJ/mol)	k^o (S cm^{-1})	R^2
BTEA: FA	38.194	2867.080	0.996
BTEA: AA	35.222	705.727	0.993
BTEA: BA	31.601	92.346	0.997
BTEA: VA	33.267	139.471	0.998
BTEA: PA	45.168	3534.684	0.992
BTEA: CAP	29.992	25.409	0.998
BTEA: CA	32.042	33.788	0.998
BTEA: LAU	31.603	17.440	0.998
BTEA: LA	48.459	34224.239	0.994
BTEA: LEV	38.931	1357.512	0.993
BTEA: OLE	34.347	31.856	0.998
BTEA: PRO	30.830	141.194	0.997
BTEA: BUT	27.109	37.800	0.991
BTEA: EG	27.574	77.644	0.996

$$\Lambda_m = \frac{\kappa M}{\rho} \tag{20}$$

The obtained linear relationship between conductivity and temperature is shown in Fig. 11b as ln n vs 1/T. The resulting value of its adjustable parameters κ° and E_a of all the prepared DESs are displayed in Table 7.

3.11. Walden plot

The Walden rule states that the following equation (21), and 22 can be used to illustrate the relationship between molar conductivity and dynamic viscosity:

$$\Lambda\eta^{\alpha} = k \tag{21}$$

$$\log\Lambda = \log k + \alpha\log\eta^{-1} \tag{22}$$

where Λ is the molar conductivity, η is dynamic viscosity, and temperature-dependent constant k is given. Fig. 12 illustrates the variation between log Λ and log η^{-1} for all DESs at a temperature ranging from 293.15 K to 343.15 K. The slope value (α) of all the synthesized DESs is represented in Table S19. The obtained Walden plot (Fig. 12) of all synthesized DESs is compared with the ideal line (resulting from 0.01 M KCl aqueous solution with slope 1). Fluid ionicity and the DESs good or evil nature is taken into consideration when analyzing deviations from this ideal line. Poor DESs are those that fall below the ideal line, and departures from the ideal line are used to characterize a decline in ionicity. As a result, Walden plots have also been considered in the literature when examining the characteristics of DESs [58]. The DESs are "subionic" because their lines are lower than the ideal KCl line [31]. The

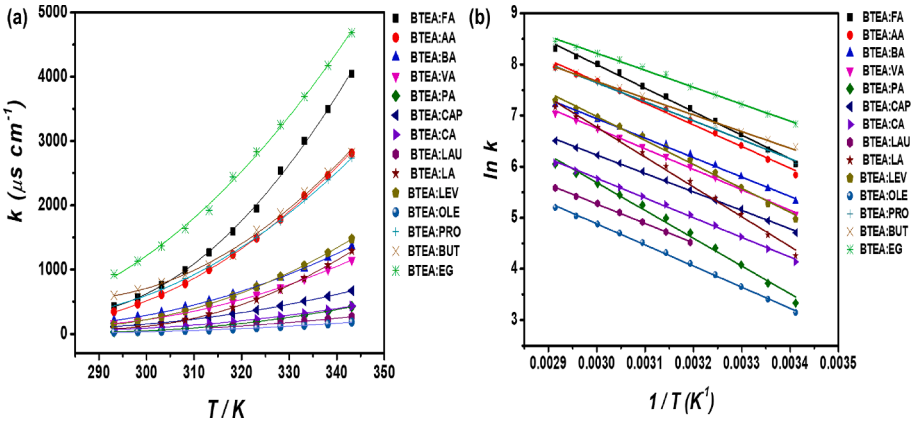


Fig. 11. (a) The plot of variation in specific conductivity of all DESs concerning temperatures, (b) Arrhenius plot (equation (19) representing the variation in conductivity, ln k, as a function of temperature.

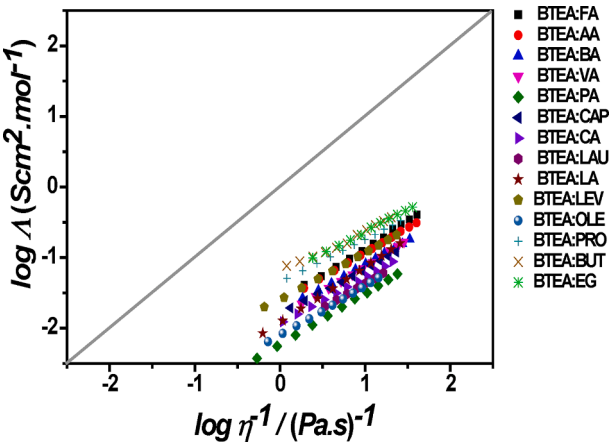


Fig. 12. The Walden plot for all synthesized DESs.

majority of ILs show smaller departures from the ideal reference line than do DESs, which may be related to the distinct migratory species present in ILs (ions) and in DESs (ions + HBDs), respectively [29]. According to the experimental results, DESs based on alcohols (BTEA: EG, BTEA: PRO, and BTEA: BUT) have similar levels of ionicity, however, DESs based on acids vary in ionicity depending on the kind of DESs (hydrophilic or hydrophobic). It was fascinating to observe that hydrophobic DESs (BTEA: OLE, BTEA: LAU, BTEA: CAP, BTEA: CA) are less ionic because they lie further from the reference line while hydrophilic DESs are more ionic since they lie closer to it.

4. Conclusions

This work effectively synthesized 14 distinct DESs (including hydrophilic as well as hydrophobic) by using benzyl triethylammonium chloride as HBA with several acids and alcohols as HBDs. Synthesized DESs are showing good thermal stability in the temperature range of 453.15–498.15 K and free-flowing ability. Physical and derived properties have been analyzed as a function of temperature and structure. It has been found that as the temperature rises viscosity decreases and conductivity increases non-linearly whereas other properties such as density, surface tension, refractive index, and sound speed show a linear downward trend. The data for molar volume, intermolecular free length, isentropic compressibility, and standard entropy showed an upward trend with temperature. No appreciable variation in hardcore molecular volumes was found with the rise in temperature whereas an increase in the free volume of DESs was observed with the temperature rise. Viscosity measurements show the viscoelastic nature of DESs. Physical properties were also found to be significantly influenced by the change in the molecular weight/chain length of components of various DESs. The surface tension of DESs containing alcohols was higher than that of DESs based on acids, which may be related to the strong hydrogen-bonding interaction between HBA and alcohols, whereas the density of benzyl tri (alkyl) ammonium chloride-based DESs having the same HBD decreases in the order methyl > ethyl > propyl > butyl and single acid-based DESs displayed a lower density value than the double acid-based DESs. Interestingly, hydrophilic DESs are denser than water while hydrophobic DESs have lower density as compared to water. Conductivity and viscosity correlation reveals that the synthesized DESs are subionic in character wherein hydrophobic DESs are less ionic as compared to hydrophilic DESs. In a nutshell, the studies will allow the selection of DESs with desired physicochemical properties in different applications and lead to further development and utilization of novel DESs as a greener alternative to organic solvents in the design and optimization of chemical and industrial processes.

CRediT authorship contribution statement

Kuldeep Singh: Conceptualization, Investigation, Data curation, Formal analysis, Methodology, Writing – original draft. **Arvind Kumar:** Conceptualization, Methodology, Resources, Supervision, Data curation, Funding acquisition, Investigation, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary material

† Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2023.122589.

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